

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-13579778})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}3291 + \mathbf{Z}(428 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 13579778 = 0$, of discriminant -54319112 . Class group invariants $(60 \ 10 \ 5)$, class number 3000; the three stored generator ideals (HNF) are $\begin{pmatrix} 1041 & 226 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2103 & 1081 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 539)y^3 + (30s + 64173)y^2 + (207s + 1473998)y + (1786s + 8130240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 1077y^8 + 129424y^7 + 16689949y^6 + 758920670y^5 \\ & + 20356771897y^4 + 397585158364y^3 + 5253242805046y^2 + 340088571 \\ & 71352y + 109417524002888 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + \left(\frac{143567}{386046315190097451} \right) s, \quad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $163263200998400 = 2^{10} \cdot 5^2 \cdot 6377468789$:

$$\begin{pmatrix} 255098751560 & 217590784500 & 81631429188 & 36234736638 & 242929058888 & 53845597857 & 204592189846 & 118214273950 & 228296114300 & 62396063166 \\ 0 & 20 & 14 & 0 & 4 & 13 & 10 & 3 & 3 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 3 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 334 factors with signed exponents of absolute value up to 5380770661691368; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 539)y^3 + (30s + 64173)y^2 + (207s + 1473998)y + (1786s + 8130240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 1077y^8 + 129424y^7 + 16689949y^6 + 758920670y^5 \\ & + 20356771897y^4 + 397585158364y^3 + 5253242805046y^2 + 340088571 \\ & 71352y + 10941752400288 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + \left(\frac{13914}{7319366412146449} \right) s, \quad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4245197219812 = 2^2 \cdot 29^2 \cdot 5189 \cdot 243197$:

$$\begin{pmatrix} 73193055514 & 60191224630 & 27811533689 & 34498423430 & 1185588565 & 45364691794 & 41037903127 & 11669585452 & 44436630101 & 39170661575 \\ 0 & 29 & 17 & 13 & 10 & 6 & 8 & 17 & 2 & 23 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 339 factors with signed exponents of absolute value up to 10589744632804631; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 539)y^3 + (30s + 64173)y^2 + (207s + 1473998)y + (1786s + 8130240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 - 1077y^8 + 129424y^7 + 16689949y^6 + 758920670y^5 \\ + 20356771897y^4 + 397585158364y^3 + 5253242805046y^2 + 340088571 \\ 71352y + 109417524002888$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + \left(\frac{12625}{2278080284401899} \right) s, \quad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2910829817600 = 2^8 \cdot 5^2 \cdot 61 \cdot 1381 \cdot 5399$:

$$\begin{pmatrix} 9096343180 & 2457015784 & 3993872520 & 3138775682 & 3129597132 & 1012554816 & 4663023730 & 6302859930 & 3281483037 & 6225601962 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 20 & 2 & 10 & 12 & 6 & 15 & 18 & 4 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 325 factors with signed exponents of absolute value up to 5943257925666664; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 1 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 6569y^3 + 12sy^2 - 320888y - 192s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 13138y^8 + 42509985y^6 + 6171314576y^4 + 40393491520y^2 + \\ 500604936192$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ -1 \ -2 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + \left(\frac{143567}{386046315190097451}\right) s, \quad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $74610669539592 = 2^3 \cdot 3^6 \cdot 101 \cdot 126666581$:

$$\begin{pmatrix} 230279844258 & 136185311382 & 63265518503 & 121542316221 & 165485779116 & 29964793632 & 147546392738 & 164733312844 & 133774902679 & 152356274247 \\ 0 & 6 & 0 & 3 & 0 & 0 & 3 & 3 & 1 & 4 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 5 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 232 factors with signed exponents of absolute value up to 4546993222; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = \begin{pmatrix} 1 & 0 & 1 \end{pmatrix}$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 6569y^3 + 12sy^2 - 320888y - 192s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 13138y^8 + 42509985y^6 + 6171314576y^4 + 40393491520y^2 + 500604936192$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ -1 \ -2 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + \left(\frac{13914}{7319366412146449}\right) s, \quad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10368548327232 = 2^6 \cdot 3^4 \cdot 131 \cdot 15267983$:

$$\begin{pmatrix} 72003807828 & 22001928450 & 16364621374 & 14955116796 & 52601643773 & 41539441693 & 35974879482 & 54114328702 & 20070849428 & 22537834490 \\ 0 & 6 & 0 & 0 & 3 & 3 & 0 & 3 & 4 & 4 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 6 & 3 & 2 & 0 & 3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 237 factors with signed exponents of absolute value up to 5236552781; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 6569y^3 + 12sy^2 - 320888y - 192s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 13138y^8 + 42509985y^6 + 6171314576y^4 + 40393491520y^2 + 500604936192$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ -1 \ -1 \ -2 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + \left(\frac{12625}{2278080284401899}\right)s, \quad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $817477551936 = 2^6 \cdot 3^6 \cdot 131^2 \cdot 1021$:

$$\begin{pmatrix} 4815036 & 3979392 & 1105116 & 4584729 & 2463288 & 1041164 & 1823211 & 3405100 & 623830 & 4342228 \\ 0 & 6 & 0 & 3 & 3 & 3 & 0 & 3 & 4 & 1 \\ 0 & 0 & 786 & 96 & 3 & 651 & 618 & 468 & 238 & 220 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 3 & 1 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 4 & 3 & 3 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 237 factors with signed exponents of absolute value up to 2773741093; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (-s - 2221)y^3 + (-33s - 102794)y^2 + (-146s - 1892002)y + (2106s - 6781636)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4442y^8 - 205588y^7 + 14728615y^6 + 1339313024y^5 + 37724 \\ & 552738y^4 + 492753650160y^3 + 3375826274932y^2 + 17310826109888y \\ & + 106220107094104 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + \left(\frac{143567}{386046315190097451} \right) s, \quad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $24746154476432 = 2^4 \cdot 71 \cdot 491 \cdot 2269 \cdot 19553$:

$$\begin{pmatrix} 3093269309554 & 1941286962798 & 1473952629112 & 2734410274934 & 2484490901202 & 1901316118702 & 1838230831685 & 2324168963343 & 402940061931 & 506566045747 \\ 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 338 factors with signed exponents of absolute value up to 5040073251; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (-s - 2221)y^3 + (-33s - 102794)y^2 + (-146s - 1892002)y + (2106s - 6781636)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4442y^8 - 205588y^7 + 14728615y^6 + 1339313024y^5 + 37724 \\ & 552738y^4 + 492753650160y^3 + 3375826274932y^2 + 17310826109888y \\ & + 106220107094104 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + \left(\frac{13914}{7319366412146449} \right) s, \quad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $141442097662363 = 141442097662363$:

$$\begin{pmatrix} 141442097662363 & 58117917105825 & 34728926932359 & 24015560517198 & 56731005413023 & 15798015091734 & 89365329646601 & 101965736699696 & 120078786569384 & 21996769926247 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 337 factors with signed exponents of absolute value up to 4355775106; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 2 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (-s - 2221)y^3 + (-33s - 102794)y^2 + (-146s - 1892002)y + (2106s - 6781636)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4442y^8 - 205588y^7 + 14728615y^6 + 1339313024y^5 + 37724 \\ & 552738y^4 + 492753650160y^3 + 3375826274932y^2 + 17310826109888y \\ & + 106220107094104 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + \left(\frac{12625}{2278080284401899} \right) s, \quad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1204916040512 = 2^6 \cdot 179 \cdot 223 \cdot 471649$:

$$\begin{pmatrix} 75307252532 & 23924063800 & 61571877590 & 48542381732 & 14637323823 & 71620640273 & 487199243 & 31186189438 & 6515926994 & 11312771897 \\ 0 & 4 & 0 & 0 & 1 & 3 & 0 & 0 & 1 & 3 \\ 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 352 factors with signed exponents of absolute value up to 6165236700; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2956y^3 + 10sy^2 - 1077941y - 1330s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 5912y^8 + 6582054y^6 + 7730764992y^4 + 800734704681y^2 + 24021269304200$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + \left(\frac{143567}{386046315190097451} \right) s, \quad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7366490174264 = 2^3 \cdot 13^2 \cdot 137 \cdot 347 \cdot 114613$:

$$\begin{pmatrix} 141663272582 & 76469067311 & 19216894240 & 96156256691 & 127578343356 & 101623736140 & 28038677140 & 61441221535 & 115269642772 & 40542442915 \\ 0 & 13 & 0 & 11 & 11 & 3 & 4 & 9 & 4 & 4 \\ 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 224 factors with signed exponents of absolute value up to 6028061447; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 4 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2956y^3 + 10sy^2 - 1077941y - 1330s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 5912y^8 + 6582054y^6 + 7730764992y^4 + 800734704681y^2 + 24021269304200$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + \left(\frac{13914}{7319366412146449} \right) s, \quad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5344480162400 = 2^5 \cdot 5^2 \cdot 19 \cdot 6317 \cdot 55661$:

$$\begin{pmatrix} 133612004060 & 114526395500 & 77803436137 & 32274998504 & 129862972768 & 17068873109 & 21964988053 & 39200442738 & 15138314674 & 55136504695 \\ 0 & 10 & 5 & 0 & 8 & 0 & 5 & 0 & 6 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 230 factors with signed exponents of absolute value up to 18013244136; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2956y^3 + 10sy^2 - 1077941y - 1330s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 5912y^8 + 6582054y^6 + 7730764992y^4 + 800734704681y^2 + 24021269304200$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + \left(\frac{12625}{2278080284401899} \right) s, \quad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a')^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $329744801214400 = 2^6 \cdot 5^2 \cdot 11^2 \cdot 13 \cdot 19 \cdot 6895657$:

$$\begin{pmatrix} 374710001380 & 167916132076 & 169712046534 & 114610612113 & 49026677559 & 178205778244 & 154540843155 & 349796955917 & 195995086449 & 230267797779 \\ 0 & 22 & 18 & 20 & 20 & 12 & 10 & 0 & 12 & 6 \\ 0 & 0 & 10 & 7 & 5 & 5 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 224 factors with signed exponents of absolute value up to 10037584754; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 3 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 1869)y^3 + (-26s - 33815)y^2 + (-203s - 99010)y + (-472s - 155068)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6y^9 - 3729y^8 - 56416y^7 + 17077809y^6 + 832832850y^5 + 16207803809y^4 + 163443137484y^3 + 913201371774y^2 + 2633026583056y + 3049403346576$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + \left(\frac{143567}{386046315190097451}\right) s, \quad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1154658115584 = 2^{13} \cdot 3^5 \cdot 211 \cdot 2749$:

$$\begin{pmatrix} 41762808 & 38604348 & 6459228 & 39647408 & 4110197 & 7926864 & 39261976 & 7908084 & 23893939 & 27785384 \\ 0 & 36 & 24 & 12 & 5 & 32 & 12 & 24 & 0 & 9 \\ 0 & 0 & 12 & 6 & 4 & 4 & 0 & 6 & 9 & 7 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 4 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 2 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 174 factors with signed exponents of absolute value up to 1351530573; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (4 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 1869)y^3 + (-26s - 33815)y^2 + (-203s - 99010)y + (-472s - 155068)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6y^9 - 3729y^8 - 56416y^7 + 17077809y^6 + 832832850y^5 + 16207803809y^4 + 163443137484y^3 + 913201371774y^2 + 2633026583056y + 3049403346576$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + \left(\frac{13914}{7319366412146449}\right) s, \quad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $183602455066368 = 2^8 \cdot 3^4 \cdot 13 \cdot 681098851$:

$$\begin{pmatrix} 212502841512 & 193104458724 & 54079538970 & 134308030492 & 80242678140 & 84668171960 & 161172111958 & 199038299284 & 163789615379 & 42356719491 \\ 0 & 12 & 9 & 1 & 6 & 6 & 0 & 0 & 10 & 7 \\ 0 & 0 & 3 & 2 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 6 & 2 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 170 factors with signed exponents of absolute value up to 1809898305; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = \begin{pmatrix} 4 & 2 & 1 \end{pmatrix}$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $\begin{pmatrix} 1 & 0 & 1 \end{pmatrix}$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 1869)y^3 + (-26s - 33815)y^2 + (-203s - 99010)y + (-472s - 155068)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 3729y^8 - 56416y^7 + 17077809y^6 + 832832850y^5 + \\ & 16207803809y^4 + 163443137484y^3 + 913201371774y^2 + 26330265830 \\ & 56y + 3049403346576 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + \left(\frac{12625}{2278080284401899} \right) s, \quad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3060658886400 = 2^8 \cdot 3^2 \cdot 5^2 \cdot 29 \cdot 1832291$:

$$\begin{pmatrix} 12752745360 & 9460680000 & 7945666181 & 6507088588 & 4471326206 & 9858787858 & 11402342686 & 11988474584 & 807606403 & 5967524832 \\ 0 & 240 & 192 & 95 & 17 & 3 & 208 & 195 & 15 & 173 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 170 factors with signed exponents of absolute value up to 1099290908; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4467y^3 + 6sy^2 + 455672y - 8624s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 8934y^8 + 20865433y^6 - 3582101640y^4 - 1197707094080y^2 + 1009973935190528$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + \left(\frac{143567}{386046315190097451} \right) s, \quad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $454801834496 = 2^9 \cdot 17 \cdot 52252049$:

$$\begin{pmatrix} 14212557328 & 6325238416 & 2173077432 & 4037013851 & 5732123998 & 1728825516 & 3105346657 & 1476234930 & 705963193 & 9248375407 \\ 0 & 16 & 0 & 15 & 6 & 0 & 12 & 0 & 4 & 2 \\ 0 & 0 & 2 & 1 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 226 factors with signed exponents of absolute value up to 389270421294; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4467y^3 + 6sy^2 + 455672y - 8624s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 8934y^8 + 20865433y^6 - 3582101640y^4 - 1197707094080y^2 + 1009973935190528$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + \left(\frac{13914}{7319366412146449}\right)s, \quad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $154514518168 = 2^3 \cdot 17^5 \cdot 61 \cdot 223$:

$$\begin{pmatrix} 133663078 & 132677520 & 72787948 & 64228857 & 9538920 & 46995880 & 62054610 & 126562047 & 35847277 & 2882087 \\ 0 & 34 & 0 & 28 & 16 & 30 & 28 & 10 & 18 & 0 \\ 0 & 0 & 34 & 27 & 21 & 9 & 16 & 21 & 7 & 32 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 209 factors with signed exponents of absolute value up to 102535353822; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (0 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4467y^3 + 6sy^2 + 455672y - 8624s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 8934y^8 + 20865433y^6 - 3582101640y^4 - 1197707094080y^2 + 1009973935190528$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + \left(\frac{12625}{2278080284401899} \right) s, \quad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $79670475084800 = 2^{10} \cdot 5^2 \cdot 17 \cdot 19^2 \cdot 507109$:

$$\begin{pmatrix} 6551848280 & 6051694712 & 5228870644 & 501208839 & 6283881259 & 4778583955 & 4292076324 & 3192093377 & 4063646838 & 1842027053 \\ 0 & 38 & 30 & 17 & 17 & 5 & 15 & 35 & 29 & 28 \\ 0 & 0 & 4 & 1 & 1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 1 & 3 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 & 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 209 factors with signed exponents of absolute value up to 133583081128; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.